



基础化学 (II)

化学动力学基础-2

王 帅

2024年3月26日

化学速率描述的基本内容

1. 定义反应速率

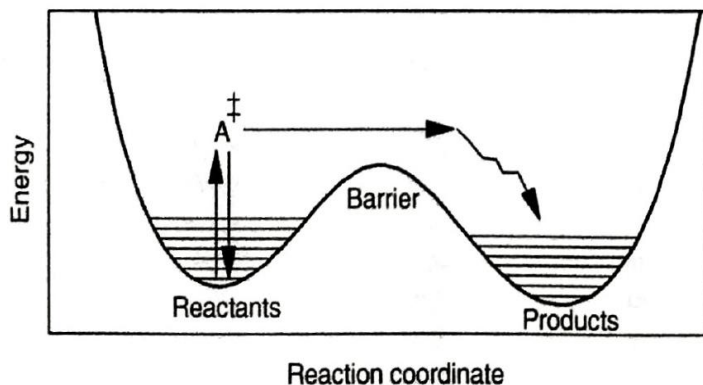
$$r = \frac{-1}{\nu_A} \frac{d[A]}{dt}$$

2. 反应速率方程

$$r = k [A]^a [B]^b \dots$$

3. 温度对速率常数的影响

$$k = A \exp\left(\frac{-E_a}{RT}\right)$$



只有**热分子**可以反应，其数量随温度升高而增多，于是反应速率随温度上升。

化学动力学提纲

1. 化学动力学引言
2. 反应速率的描述 (是什么?)
3. 反应速率理论 (为什么?)
4. 动力学实验测定 (如何做?)
5. 复杂反应动力学分析 (如何用?)

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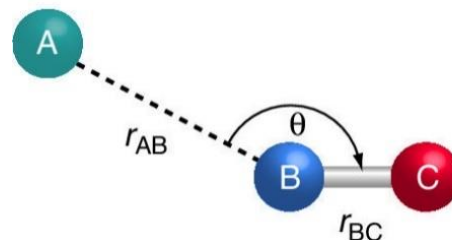


本次课内容

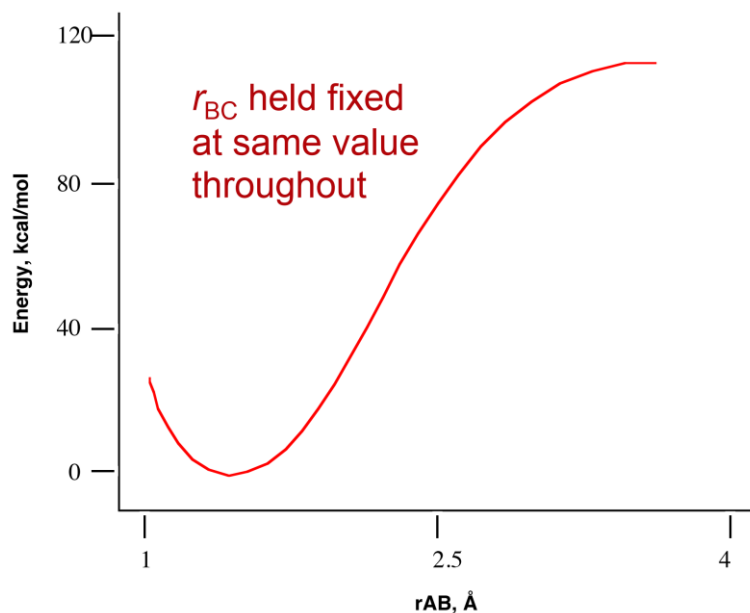
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化学反应势能面

考虑一个简单的例子



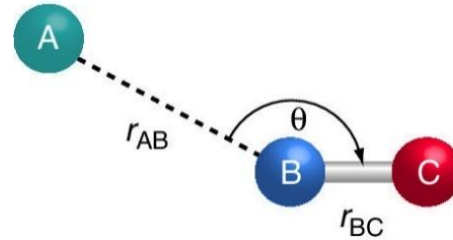
一维变量描述的反应路径



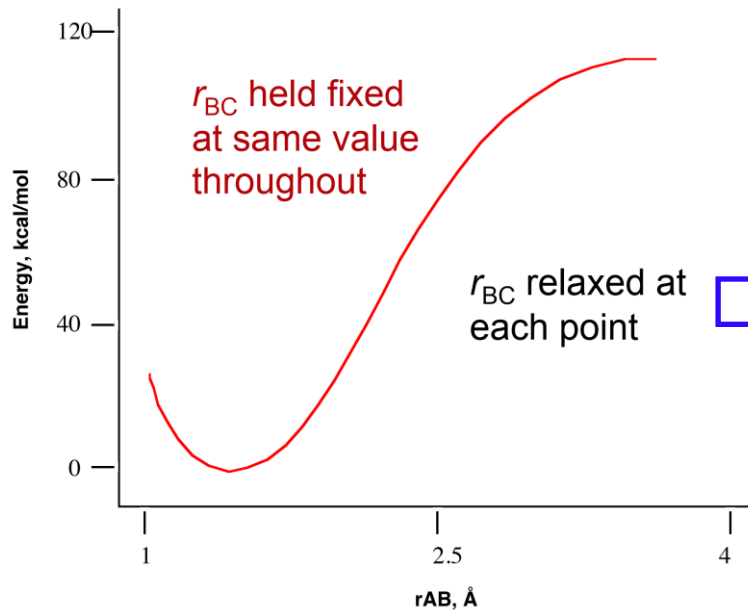
从反应物到产物的转化过程中，A-B间距离 r_{AB} 减少到A-B平衡键长，而B-C间距离 r_{BC} 增加，断键并释放C原子。A原子可从任意角 θ 进攻。

化学反应势能面

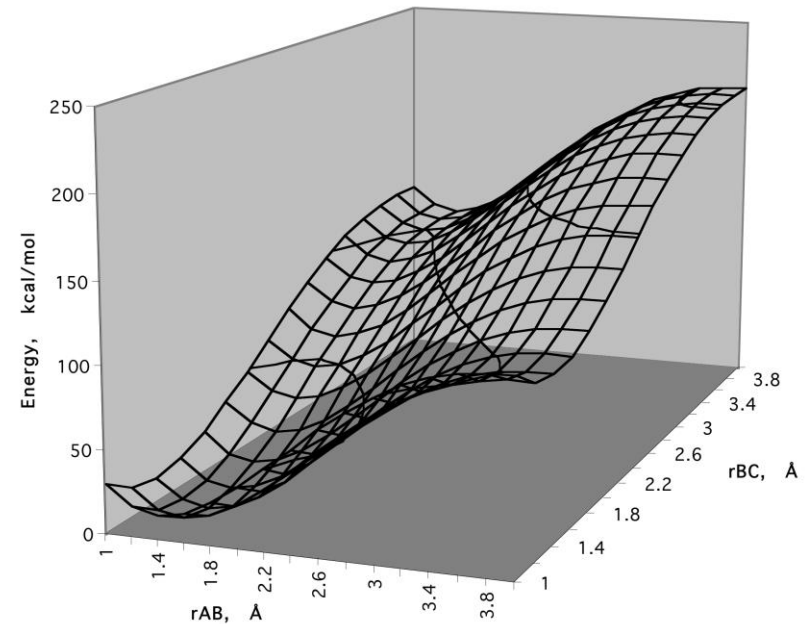
考虑一个简单的例子



一维变量描述的反应路径

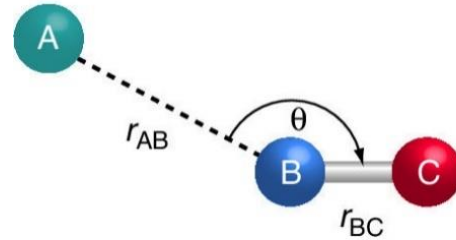


二维变量描述的反应路径



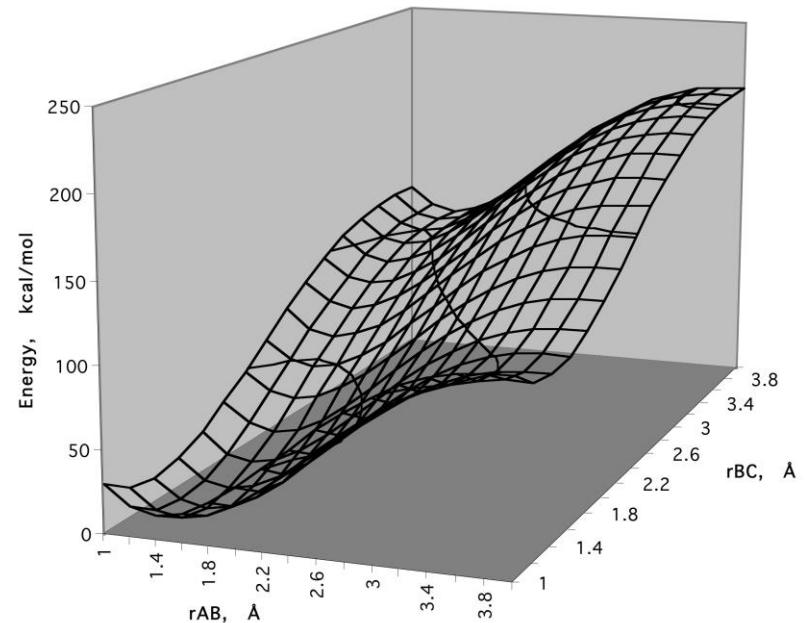
化学反应势能面

考虑一个简单的例子



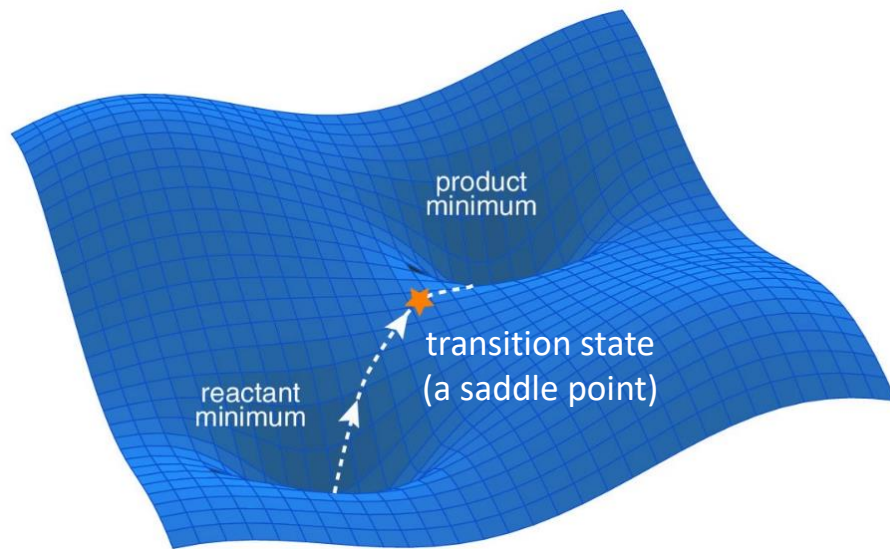
二维变量描述的反应路径

原则上，我们可以计算任意A与B原子反应过程中几何构型的能量，由此我们会得到势能面 (Potential Energy Surface)。这是一种为该系统所有原子的位置给出能量的函数。

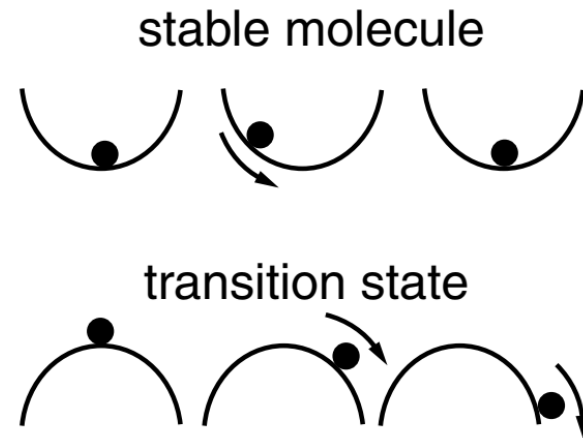


反应路径与过渡态

从反应物到产物的许多可能路径中，克服最小能垒的路径具有最大的发生概率，即**反应坐标**。该路径上的最高能量点被称为**过渡态**。在过渡态中，往往存在新键的部分生成和旧键的部分断裂。



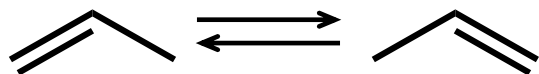
势能面与反应路径示意图



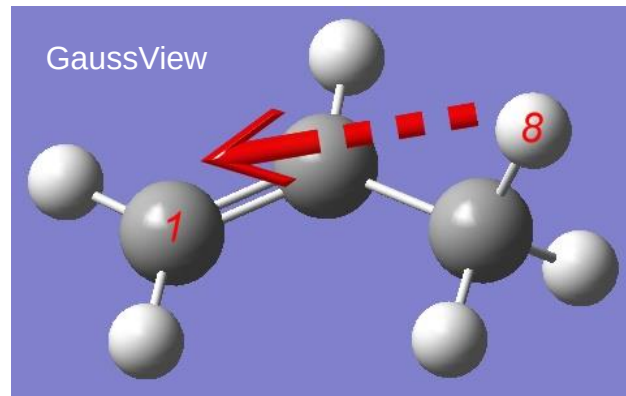
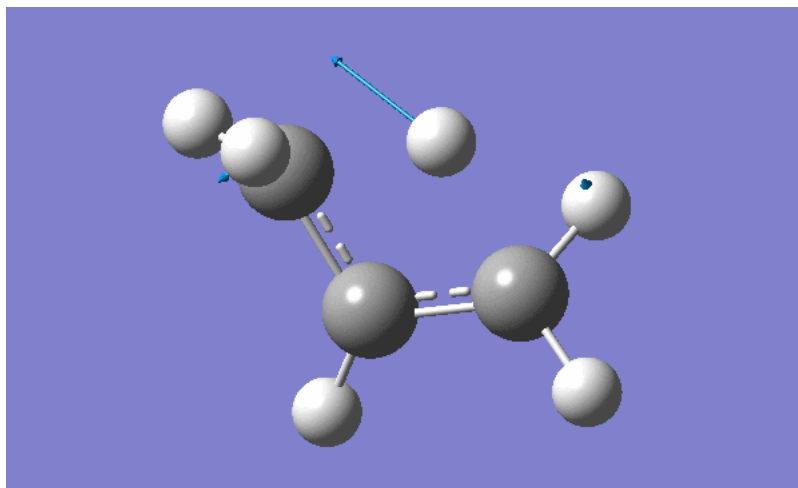
不同于稳态分子，过渡态受微扰后极易转化为反应物或产物。

反应路径与过渡态

利用Gaussian寻找丙烯双键异构的过渡态 (TS) :



过渡态结构



G5:M1:V1 - Display Vibrations

Mode #	Freq	Infrared
1	-1860.91	27.7175
2	569.28	0.8503
3	603.66	8.7129
4	842.21	87.6459
5	853.88	88.2907
6	932.53	0.5408
7	943.78	11.9140
8	1017.36	0.2007
9	1022.06	9.3288
10	1170.05	58.9020

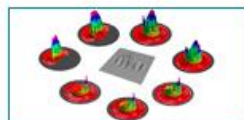
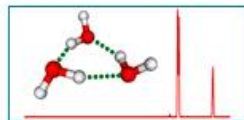
过
渡
态
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:
虚
频

Animate Vibration:
Start Animation Save Movie...
Animation Frequency:
Displacement Amplitude:

反应路径与过渡态



1111基元反应动力学研究组



头条

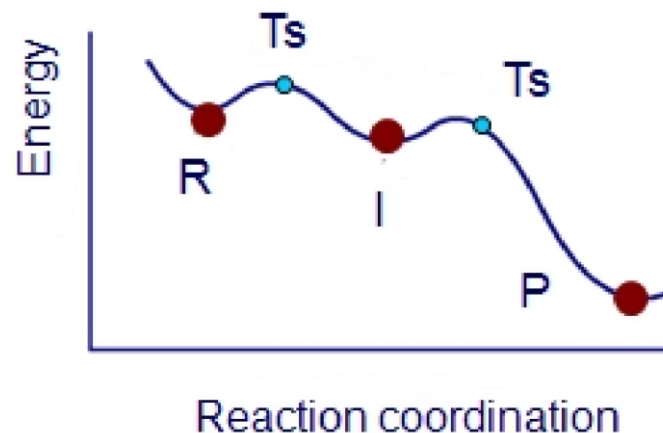
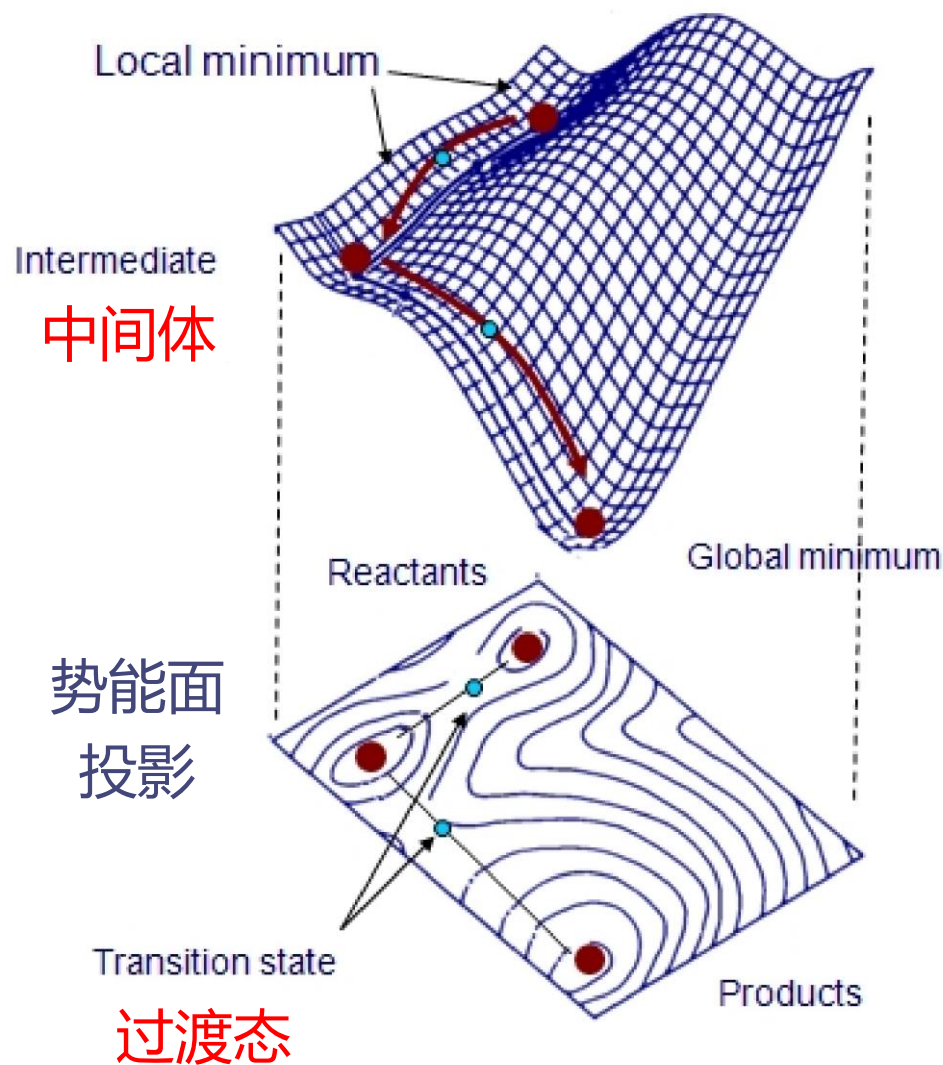
More>>

化学动力学研究中心举行研究生奖颁奖仪式

化学动力学研究中心于2017年获得国家自然科学基金委科学中心“动态化学前沿研究”的项目支持。根据基金委科学中心项目的精神，为确保研究工作的顺利开展，完满完成项目目标，设立了“化学动...>详细内容

“杨学明院士：亲手研制仪器，破解科学难题”——2002年回国至今，已发表15+篇Science、2+篇Nature论文，两获国家自然科学基金二等奖。

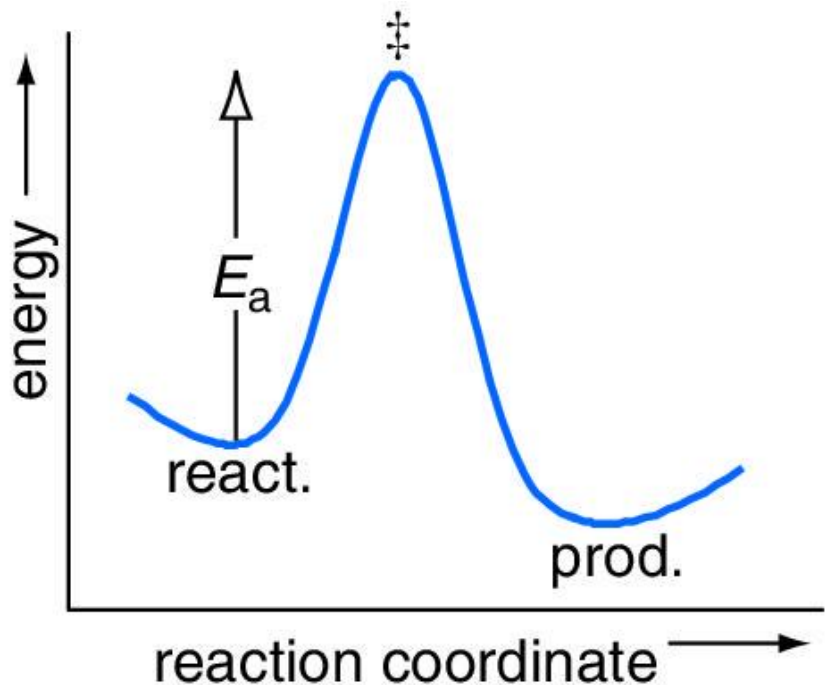
过渡态 vs. 中间体



中间体：如其他稳态分子一样，处于势能面的极小值处，并可被常规方法检测。它可能比反应物与产物能量更高。

过渡态：处于势能面的极大值处，只具有非常短的寿命 ($\sim 10^{-13}$ s)，需要借助飞秒激光技术检测。

过渡态与活化能

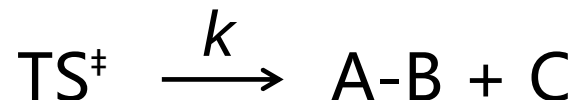
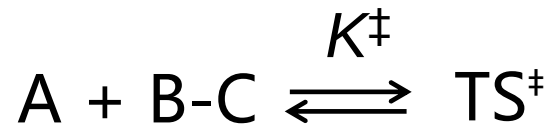
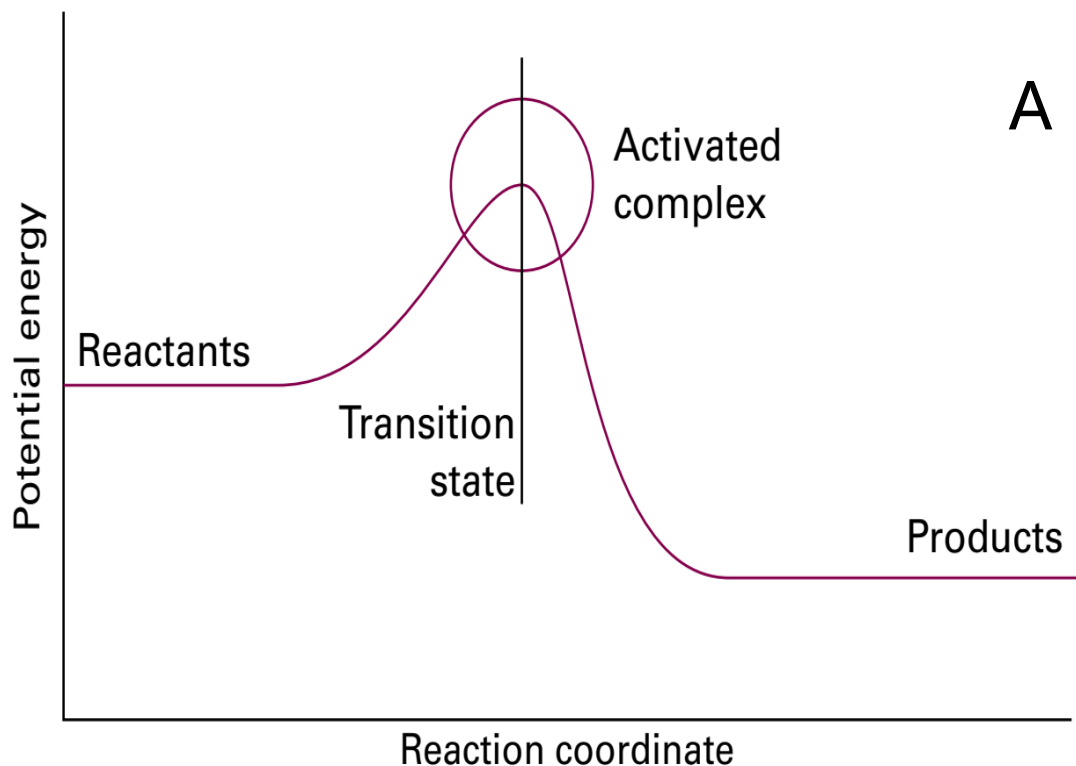


将能量与反应坐标作图，得到反应能量图(the energy profile of the reaction)。其中，过渡态(用 $\ddagger$ 表示)与反应物的能量之差即为活化能(E_a)，决定反应速率。

根据Boltzmann分布，分子中含有足够能量达到过渡态的比例是 $\exp(-E_a/RT)$ 。典型的活化能值大约为 50 kJ mol^{-1} ，即反应分子只有约 $1/10^9$ 概率爬过能垒，转化为产物。

经典过渡态理论

过渡态理论，又常被称为活化络合物理论或绝对反应速率理论，是以假定过渡态与反应物电子态之间存在宏观时间无法分辨的化学平衡为出发点。



$$\text{rate} = k [TS^\ddagger]$$

$$= k \cdot K^\ddagger [A][B-C]$$

$$= k_{\text{obs}} [A][B-C]$$

经典过渡态理论

经典过渡态理论的基本假设：

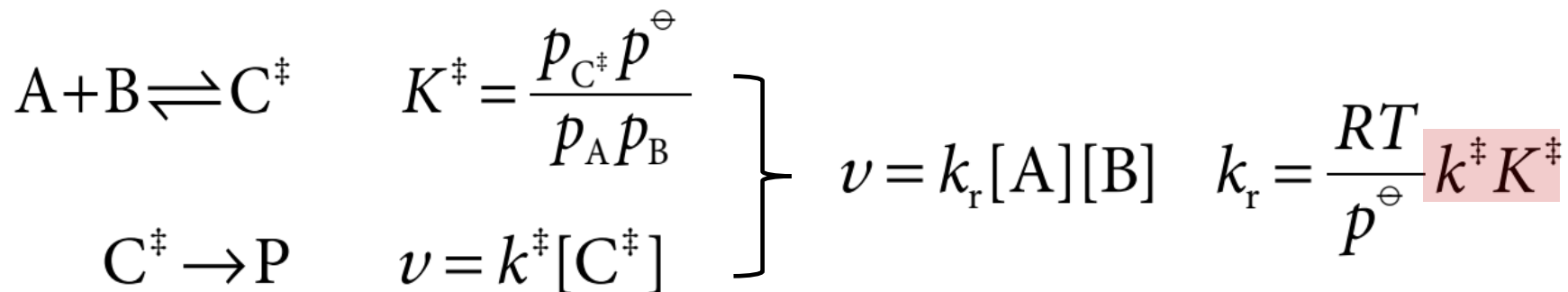
- (1) 反应分子应先活化为处于过渡态的活化络合物，然后再单方向地分解为产物，即过渡态是自反应物向产物过渡的一个不折回点，决定反应速率 (**过渡态假设**)。
- (2) 反应物与活化络合物之间处于热平衡态，无论体系中是否存在产物以及其是否与反应物达成平衡；同时，若果无论是反应物或过渡态，均存在与体系温度相对应的 Boltzmann 分布 (**准平衡假设**)。

一般而言，若 $E_a/RT \geq 5$ ，分子间的传能速率远大于反应速率，其可使反应物服从 Boltzmann 分布这一条件得以成立。

经典过渡态理论

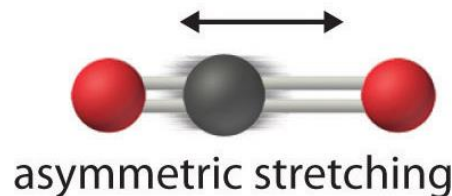
对于气相双分子基元反应：

Atkins' Physical Chemistry 11th



设 $\nu^\ddagger$ 为活化络合物沿反应路径方向的不对称伸缩振动的频率，其正比于活化络合物分解的速率常数，而引入传递系数 κ 作为比例系数(通常设为 $\kappa = 1$)。于是有，

$$k^\ddagger = \kappa \nu^\ddagger$$



经典过渡态理论

对于 $K^\ddagger$ 的处理，将活化络合物沿反应路径方向的“不可逆”化学键振动贡献 ($\approx kT/h\nu^\ddagger$) 进行分离：

$$K^\ddagger = \frac{kT}{h\nu^\ddagger} \bar{K}^\ddagger \quad \leftarrow \begin{array}{l} \text{可用统计热力} \\ \text{学严格求解} \end{array}$$

于是有：

$$\begin{aligned} k_r &= \frac{RT}{p^\ominus} k^\ddagger K^\ddagger = \kappa \nu^\ddagger \frac{kT}{h\nu^\ddagger} \frac{RT}{p^\ominus} \bar{K}^\ddagger \\ &= \kappa \frac{kT}{h} \frac{RT}{p^\ominus} \bar{K}^\ddagger \quad (\text{Eyring equation}) \end{aligned}$$

若使用浓度平衡常数，易得 $k_r = \kappa \frac{kT}{hc^\ominus} \bar{K}_c^\ddagger$ (适于溶液反应)

经典过渡态理论

引入活化吉布斯自由能以及活化焓、活化熵：

$$\left. \begin{aligned} \Delta^\ddagger G &= -RT \ln \bar{K}^\ddagger \\ \Delta^\ddagger G &= \Delta^\ddagger H - T\Delta^\ddagger S \end{aligned} \right\} k_r = \kappa \frac{kT}{h} \frac{RT}{p^\ominus} e^{-\Delta^\ddagger G/RT}$$

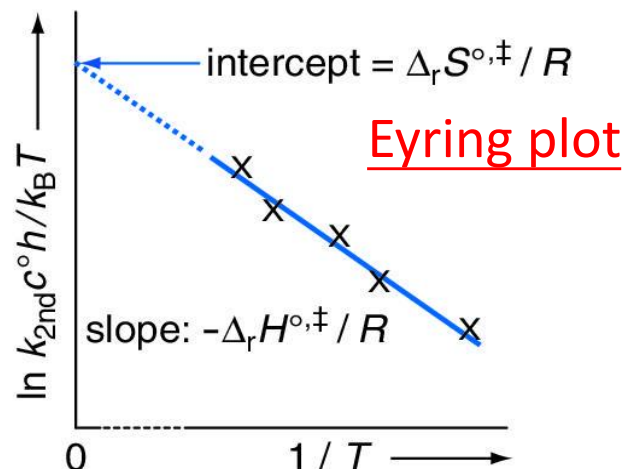
**指前因子与
活化熵相关**

$$= \frac{kT}{h} \frac{RT}{p^\ominus} e^{\Delta^\ddagger S/R} e^{-\Delta^\ddagger H/RT}$$

对于溶液体系：

$$k_{2\text{nd}} = \frac{k_B T}{c^\ominus h} \exp\left(\frac{\Delta_r S^{\circ, \ddagger}}{R}\right) \exp\left(\frac{-\Delta_r H^{\circ, \ddagger}}{RT}\right)$$

$$\ln\left(\frac{k_{2\text{nd}} c^\ominus h}{k_B T}\right) = \frac{\Delta_r S^{\circ, \ddagger}}{R} - \frac{\Delta_r H^{\circ, \ddagger}}{RT}$$



经典过渡态理论

根据实验活化能的定义 (IUPAC, 1996):

$$E_{a, \text{exp}} = RT^2 \frac{d \ln k (T)}{dT}$$

对于**气相**双原子反应,

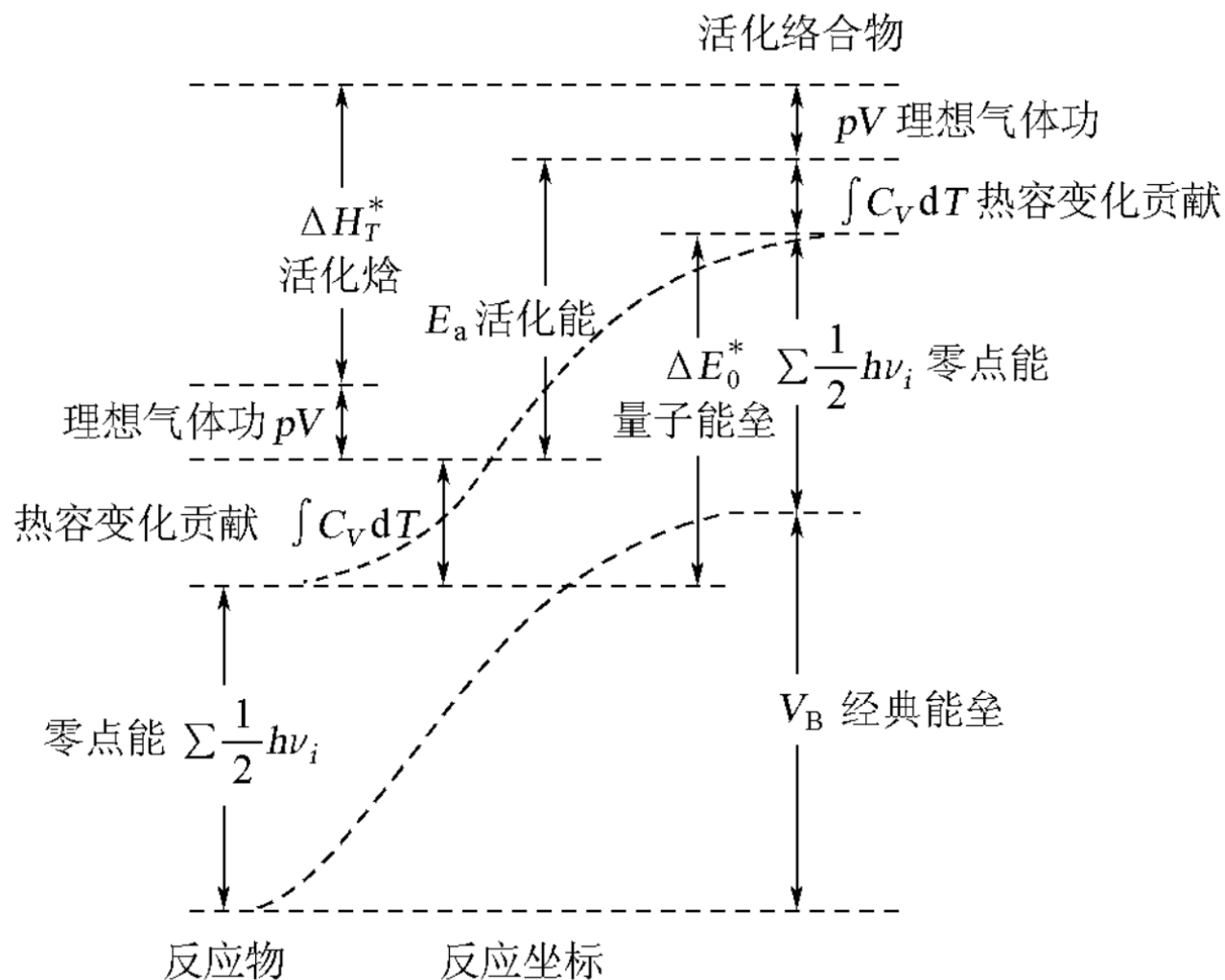
$$k_r = \kappa \frac{kT}{h} \frac{RT}{p^\ominus} \bar{K}^\ddagger \quad \Rightarrow \quad \Delta^\ddagger H = E_a - 2RT$$

对于**液相**双原子反应,

$$k_r = \kappa \frac{kT}{hc^\ominus} \bar{K}_c^\ddagger \quad \Rightarrow \quad \Delta^\ddagger H = E_a - RT$$

思考题：推导气相单原子反应活化能与活化焓的关系。

关于活化能与其他能垒



知识介绍

再谈什么是活化能

——Arrhenius 活化能的定义、解释、以及容易混淆的物理量

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化工学院 江苏南京 210093, ³北京大学化学与分子工程学院 北京 100871)

Changing Conceptions of Activation Energy

Philip D. Pacey

Dalhousie University, Halifax, Nova Scotia B3H 4J3 Canada

Pacey P D. *J Chem Educ*, 1981, 58 612

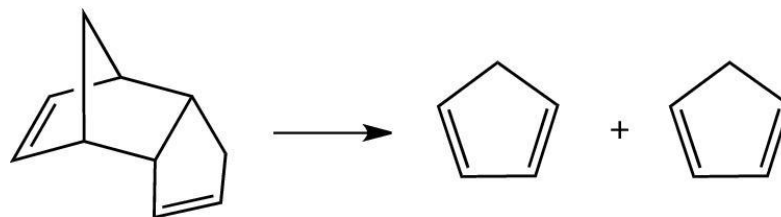
经典过渡态理论

由于活化焓与活化能十分接近，可以用相同方式来理解它们反应速率的影响。这里仅举例一些活化熵对反应速率的影响。



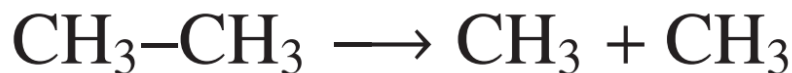
$$\Delta_r S^{\circ, \ddagger} = -150 \text{ J K}^{-1} \text{ mol}^{-1}$$

两个分子形成的过渡态失去了一些平动自由度



$$\Delta_r S^{\circ, \ddagger} = -8 \text{ J K}^{-1} \text{ mol}^{-1}$$

很小的熵变，暗示过渡态与反应物的结构更加接近



$$\Delta_r S^{\circ, \ddagger} = 58 \text{ J K}^{-1} \text{ mol}^{-1}$$

很大的熵变，暗示过渡态与产物的结构更加接近



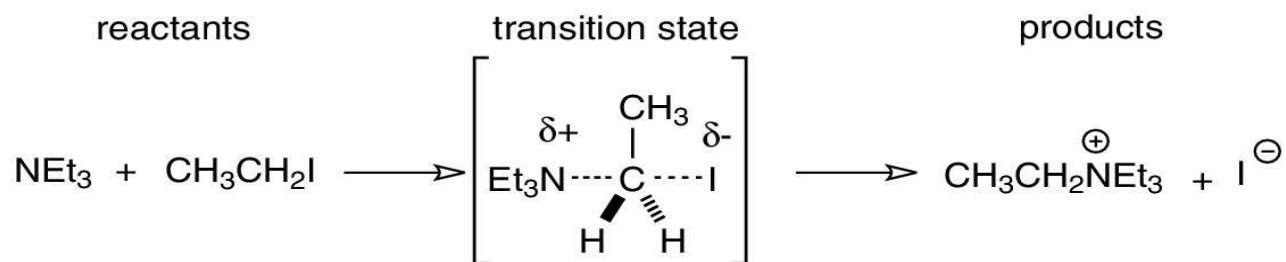
$$\Delta_r S^{\circ, \ddagger} = -32 \text{ J K}^{-1} \text{ mol}^{-1}$$

熵减小... 暗示过渡态是“扭曲”六元环状结构

活化熵的溶剂效应

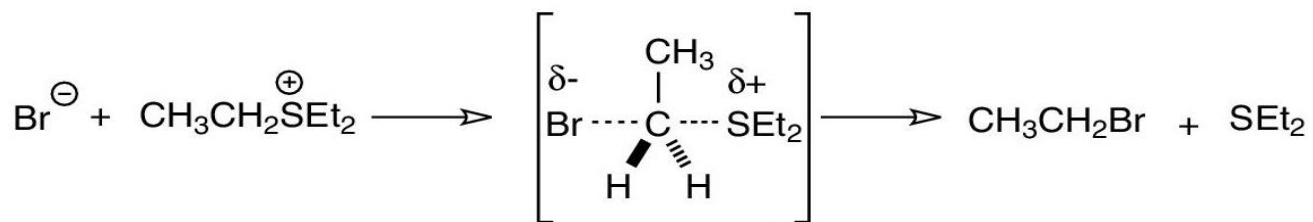
对于在溶液中的双分子反应：

(1) Reactions characterized by charge separation



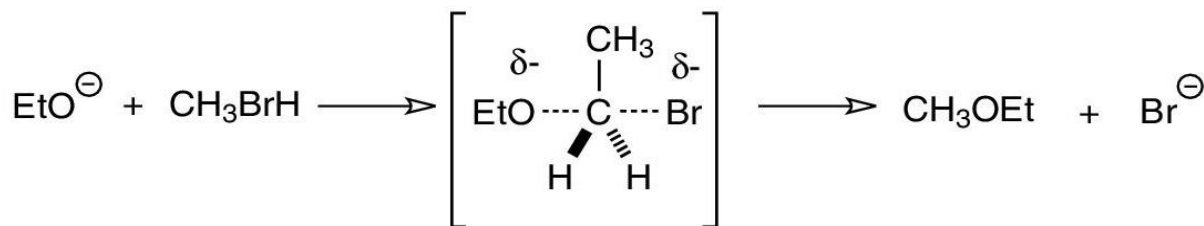
$$\Delta_r S^\circ, \ddagger \\ -115 \text{ J K}^{-1} \text{ mol}^{-1}$$

(2) Reactions characterized by charge neutralization



$$\Delta_r S^\circ, \ddagger \\ +76 \text{ J K}^{-1} \text{ mol}^{-1}$$

(3) Reactions characterized by charge delocalization

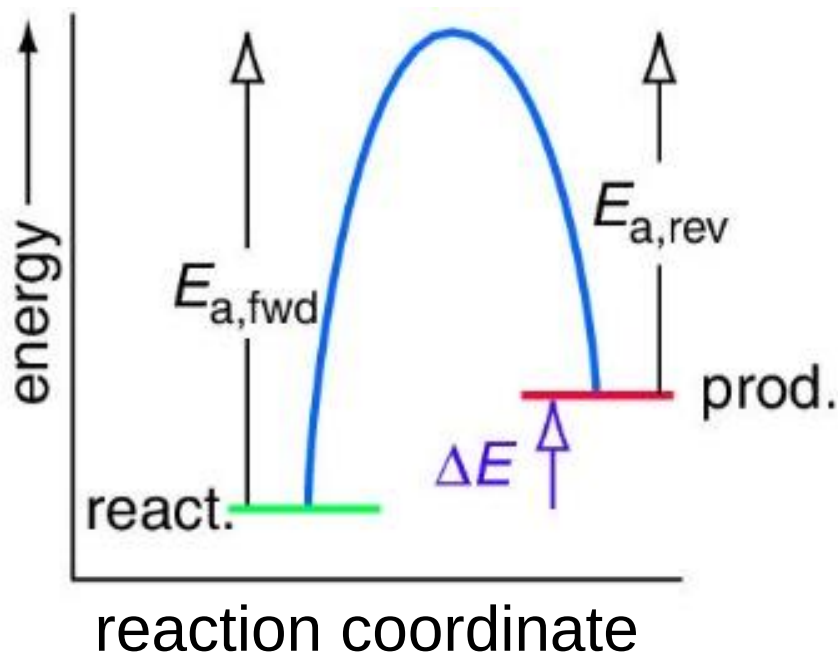


$$\Delta_r S^\circ, \ddagger \\ -26 \text{ J K}^{-1} \text{ mol}^{-1}$$

可逆反应与化学平衡

微观可逆性原理:

任意基元反应必然存在逆反应，其逆反应必然也是基元反应，而且它在正、逆方向进行时必通过同一过渡态。



一般的言，逆反应活化能 ($E_{a,\text{rev}}$) 与正反应 ($E_{a,\text{fwd}}$) 不同。它们的差别可以由反应物与产物的能量 (ΔE) 差表示:

$$\Delta E = E_{a,\text{fwd}} - E_{a,\text{rev}}$$

可逆反应与化学平衡

当一个反应达到平衡，反应物与产物的浓度都不会改变。但是，这并不表明没有反应发生。相反，它表明正反应与逆反应速率相等，即达到**动态平衡** (精细平衡原理)。

比如 $A + B \rightleftharpoons P + Q$ $K_{\text{eq}} = \frac{[P]_{\text{eq}}[Q]_{\text{eq}}}{[A]_{\text{eq}}[B]_{\text{eq}}}$

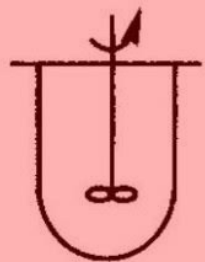
因正、逆反应速率相等 ($k_f[A]_{\text{eq}}[B]_{\text{eq}} = k_r[P]_{\text{eq}}[Q]_{\text{eq}}$)

$$K_{\text{eq}} = \frac{k_{\text{fwd}}}{k_{\text{rev}}} = \frac{A_{\text{fwd}} \exp(-E_{\text{a,fwd}}/RT)}{A_{\text{rev}} \exp(-E_{\text{a,rev}}/RT)} = \frac{A_{\text{fwd}}}{A_{\text{rev}}} \exp(-\Delta E/RT)$$

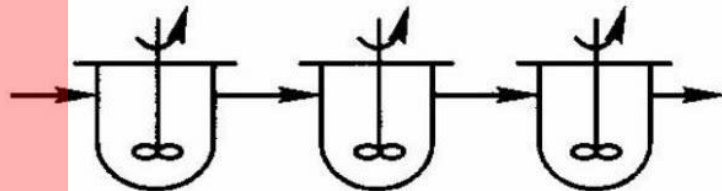
显然，平衡常数并不只取决于 ΔE ，同时与A的比值有关。

4. 动力学实验测定

典型的化工反应器



(a) 间歇搅拌釜



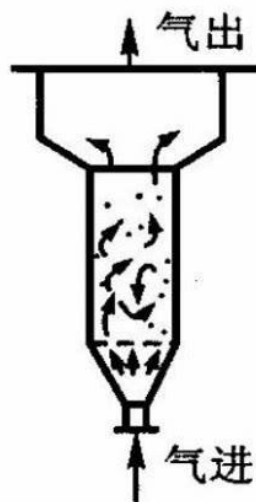
(b) 连续操作搅拌釜



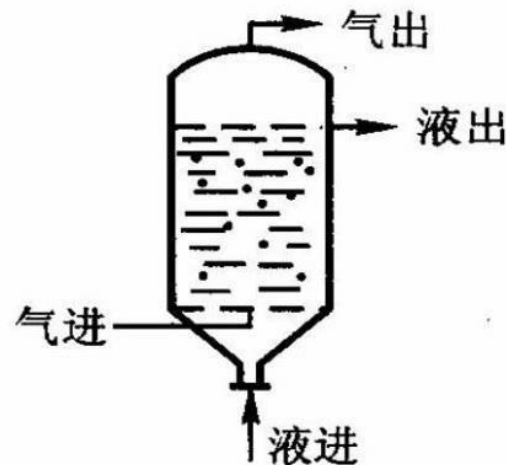
(c) 管式反应器



(d) 固定床反应器



(e) 流化床反应器



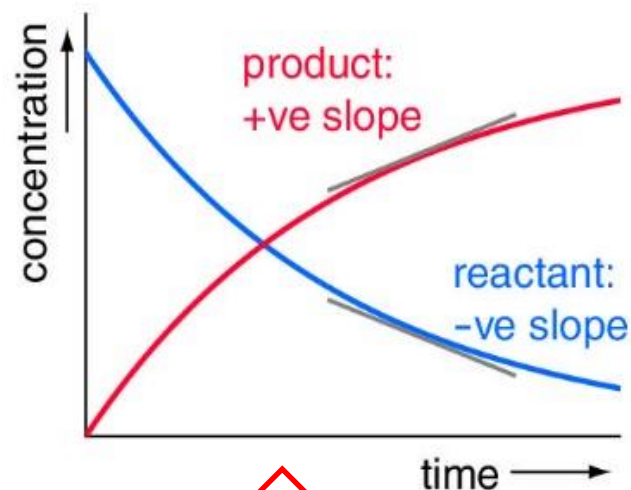
(f) 鼓泡床反应器

反应动力学分析

在本质上，反应动力学实验是定量测量反应物/产物浓度随反应时间的变化函数，并借助反应动力学模型对速率/浓度变化的模拟来推测反应机理。

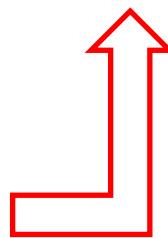


测什么?



动力学方程
是以微分形
式来定义

$$\frac{d[A]}{dt} = -k [A]^a [B]^b \dots$$



需积分方程
来拟合数据

一级反应动力学

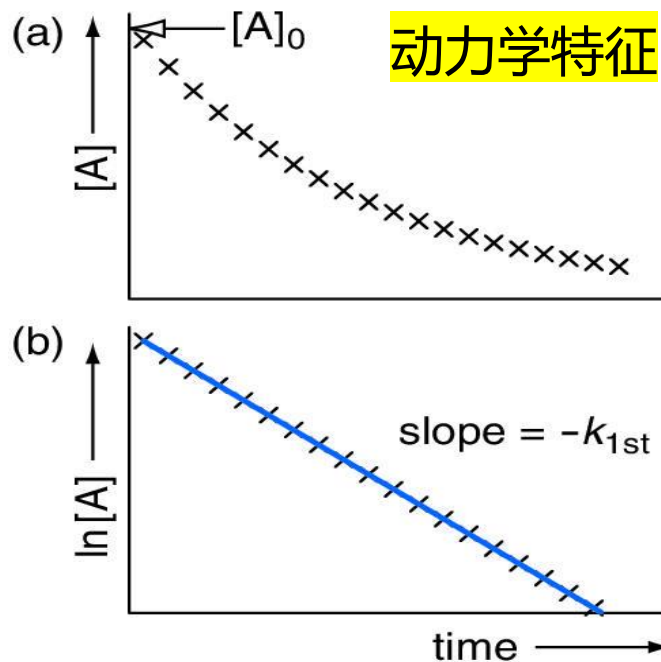
对于一级反应 $A \rightarrow B$, 其动力学方程的形式为:

$$\frac{d[A]}{dt} = -k_{1st}[A] \quad \Rightarrow \quad \frac{1}{[A]} d[A] = -k_{1st} dt$$

对方程两侧进行积分得,

$$\ln[A] = -k_{1st}t + \ln[A]_0$$

$$[A] = [A]_0 \exp(-k_{1st}t)$$



一级反应动力学

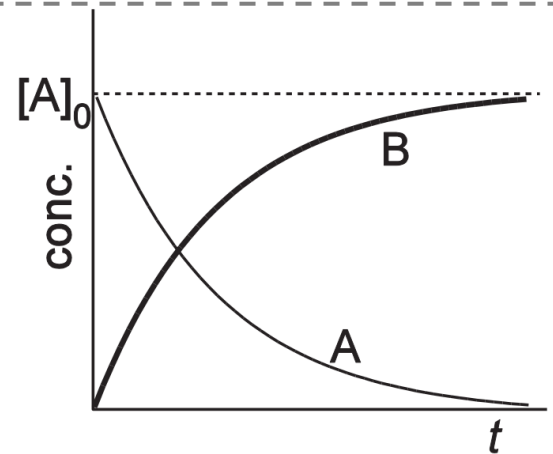
对于一级反应 $A \rightarrow B$, 以产物B的生成为研究对象:

$$\frac{d[B]}{dt} = k_{1st}[A]$$

根据如下关系易得,

$$\left\{ \begin{array}{l} [B] = [A]_0 - [A] \\ [A] = [A]_0 \exp(-k_{1st}t) \end{array} \right.$$

$$\Rightarrow [B] = [A]_0(1 - \exp(-k_{1st}t))$$



Growth of the product [B] and decay of the reactant [A] as a function of time. At all times $[A] + [B] = [A]_0$.

二级反应动力学

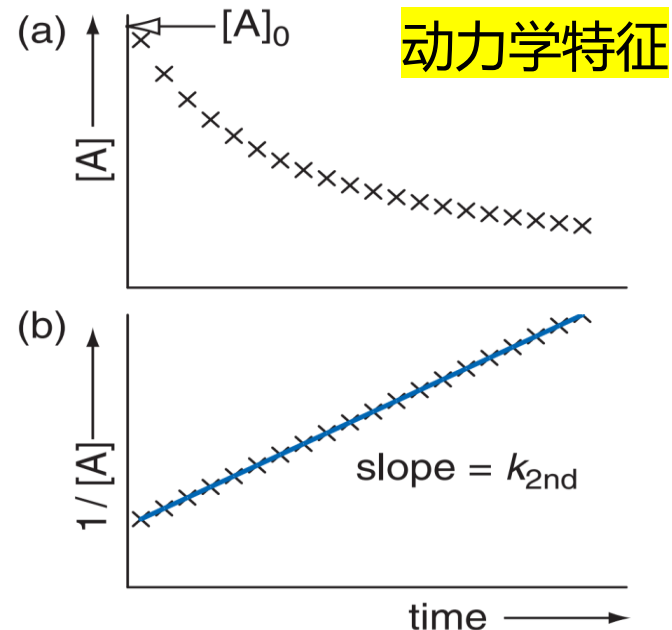
对于二级反应 $2A \rightarrow P$, 其动力学方程的形式为:

$$\frac{d[A]}{dt} = -k_{2nd}[A]^2 \Rightarrow \frac{1}{[A]^2} d[A] = -k_{2nd} dt$$

对方程两侧进行积分得,

$$\frac{1}{[A](t)} - \frac{1}{[A]_0} = k_{2nd} t$$

$$[A](t) = \frac{[A]_0}{k_{2nd} t [A]_0 + 1}$$



二级反应动力学

对于二级反应 $A + B \rightarrow P$, 其动力学方程的形式为:

$$\frac{d[A]}{dt} = -k_{2nd}[A][B]$$

在实验上通常使用两种方式简化研究, 即

(1) 使初始浓度 $[B]_0 = [A]_0$, 则在反应过程中 $[B]$ 与 $[A]$ 等量变化,

$$\frac{d[A]}{dt} = -k_{2nd}[A][B] = -k_{2nd}[A]^2$$

(2) 使初始浓度 $[B]_0 \gg [A]_0$, 则在反应过程中 $[B]$ 近似不变,

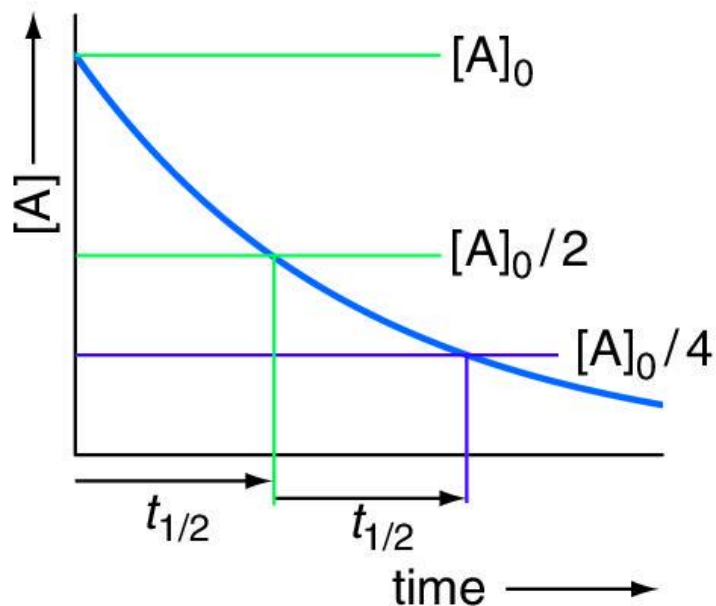
$$\frac{d[A]}{dt} = -k_{2nd}[B]_0[A] = -k'_{1st}[A]$$

分离变量法
的缺点?

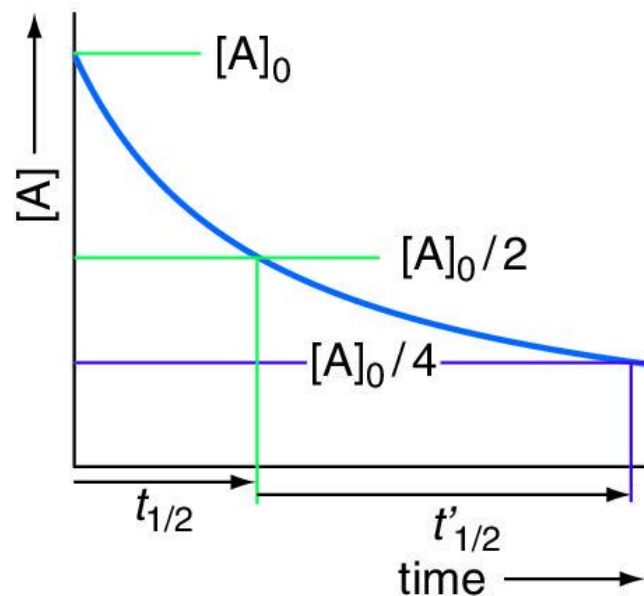
化学反应的半衰期

半衰期 (Half-life) 是指某种特定物质的浓度经过某种反应降低到剩下初始时一半所消耗的时间，是研究反应动力学的一个容易测定的重要参数。

(a) 一级反应 $t_{1/2} = \frac{1}{k_{1st}} \ln 2$



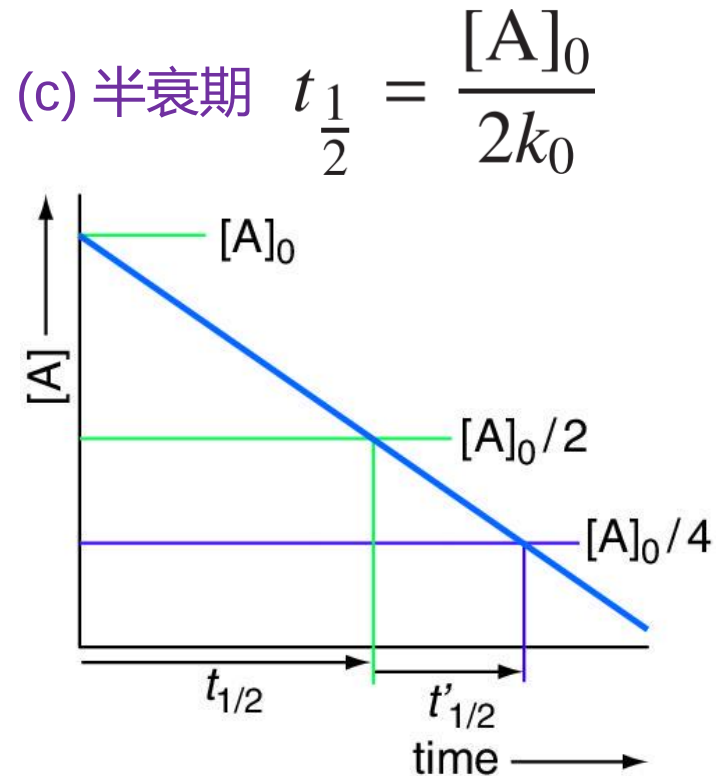
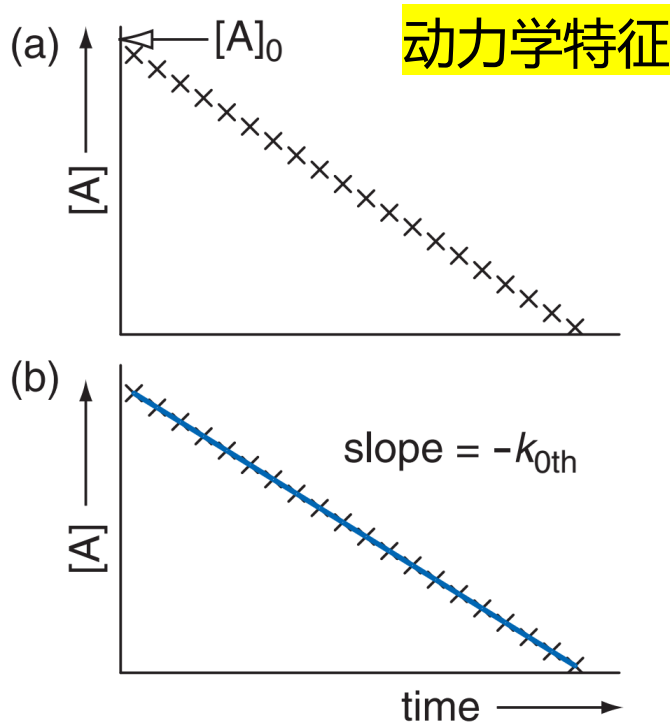
(b) 二级反应 $t_{1/2} = \frac{1}{k_{2nd}} \frac{1}{[A]_0}$



零级反应动力学

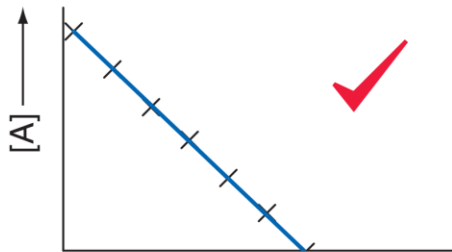
若反应速率不受其中反应物A浓度的影响, 即反应级数为零:

$$\frac{d[A]}{dt} = -k_0 \Rightarrow [A](t) = [A]_0 - k_0 t$$

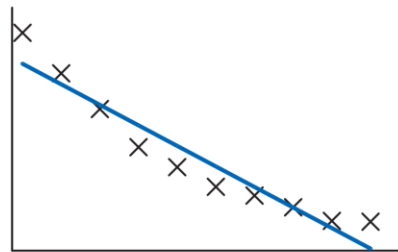


反应级数的辨别

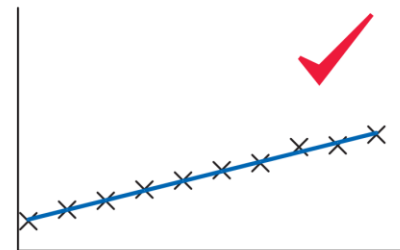
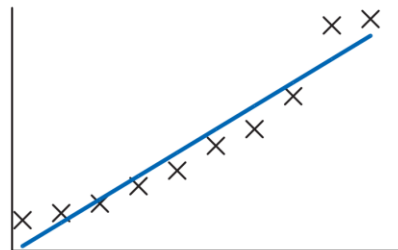
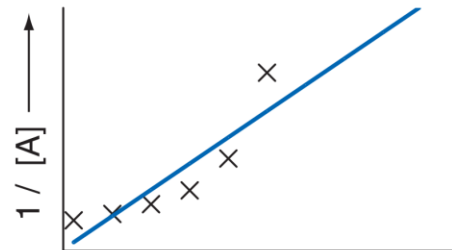
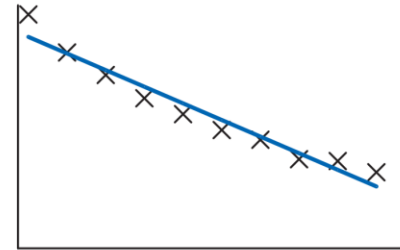
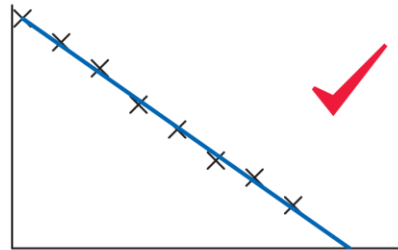
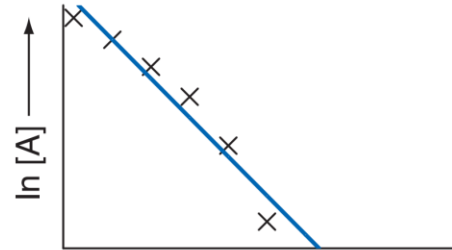
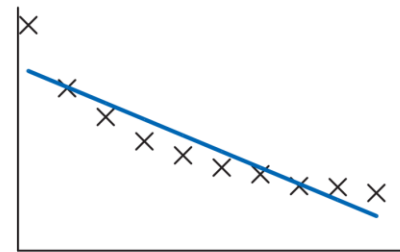
zero-order data



first-order data



second-order data



time →

1. 需要足够质量和数量的实验数据来辨别反应级数;

2. 排除一种反应级数的可能性往往比确定一种反应级数要容易得多。

化学动力学提纲

1. 化学动力学引言
2. 反应速率的描述 (是什么?)
3. 反应速率理论 (为什么?)
4. 动力学实验测定 (如何做?)
5. 复杂反应动力学分析 (如何用?)

本周作业（4道题）

- 20.1 The alkaline hydrolysis of ethyl ethanoate in aqueous solution is first order in ester and OH^- , and so second order overall

$$\frac{d[\text{OH}^-]}{dt} = -k_{2\text{nd}}[\text{ester}][\text{OH}^-].$$

In an experiment to verify this rate law, a reaction mixture was prepared in which the initial concentration of ester and OH^- were both $0.025 \text{ mol dm}^{-3}$. At intervals after the reaction was initiated, a 10 cm^3 aliquot of the reaction mixture was withdrawn and mixed with 10 cm^3 of $0.05 \text{ mol dm}^{-3} \text{ HCl(aq)}$. This amount of acid is sufficient to neutralize any unreacted OH^- , thus quenching the reaction. The remaining acid was then titrated with $0.02 \text{ mol dm}^{-3} \text{ NaOH(aq)}$ to the end point in the usual way. This method gave the following data at 0°C

t / s	vol. of titre / cm^3	$n_{\text{titre}} / \text{mol}$	$n_{\text{aliquot}} / \text{mol}$	$[\text{OH}^-] / \text{mol dm}^{-3}$
120	13.4	2.68×10^{-4}	2.32×10^{-4}	0.0232
300	14.3			
600	15.6			
900	16.4			
1200	17.4			
1500	18.1			
1800	18.5			

Complete the table in the following way (to help you, the first line has been completed).

- Work out the amount in moles of OH^- in the volume of NaOH solution added in the titration. Enter this in the column headed n_{titre} .
- Work out the amount in moles of H^+ in the acid used to quench the reaction mixture, and hence the amount in moles of *unreacted* OH^- in the aliquot: this is n_{aliquot} .
- Finally, work out the concentration of OH^- in the aliquot, and enter this into the final column.

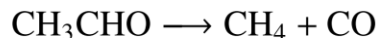
Since the initial concentrations of OH^- and ester are equal, the rate law can be simplified to $d[\text{OH}^-]/dt = -k_{2\text{nd}}[\text{OH}^-]^2$. It therefore follows that a plot of $1/[\text{OH}^-]$ against t should be a straight line.

Make such a plot and use it to estimate a value for the second-order rate constant, stating the units of your result. (These are real data, and so some scatter is to be expected).

Test the data to see if they fit a first-order rate law (i.e. plot $\ln[\text{OH}^-]$ against t).

本周作业（4道题）

20.8 The gas phase decomposition of ethanal

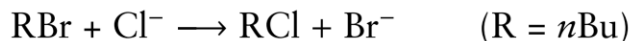


has the rate law

$$\frac{d[\text{CH}_3\text{CHO}]}{dt} = -k [\text{CH}_3\text{CHO}]^{\frac{3}{2}}.$$

- (a) Integrate this rate law and hence obtain an expression for how the concentration of ethanal varies with time (assume that the concentration of ethanal at time zero is $[\text{CH}_3\text{CHO}]_0$).
- (b) Determine the form a plot which would be expected to give a straight line for data which follows this rate law. How is the rate constant related to the slope of this plot?
- (c) Find an expression for the half life of the reaction.

20.10 Rate constants for the following reaction (in solution)



as a function of temperature are given in the following table. Use a graphical method to determine the enthalpy and entropy of activation.

$T / ^\circ\text{C}$	25.0	34.6	44.5	55.2	64.8
$10^5 \times k_{2\text{nd}} / \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$	6.45	16.4	41.0	106	215

Is your value of $\Delta_r S^\ddagger$ consistent with the reaction proceeding via a mechanism in which RBr and Cl^- come together in the rate-limiting step?

本周作业（4道题）

20.12 At high temperatures *t*-butyl chloride is pyrolysed in the gas phase to give isobutene and HCl



The progress of the reaction can be monitored by following the increase in the total pressure, p_{tot} .

- (a) If the initial pressure is $p_{\text{tot,init}}$, and is due entirely to *t*-butyl chloride, show that the partial pressure of *t*-butyl chloride is given by

$$p_{\text{BuCl}} = 2p_{\text{tot,init}} - p_{\text{tot}}$$

[Hint: use the same approach as in section 20.1.4 on page 765].

At 330 °C the following data were obtained

t / s	60	120	180	240	300	360	420	480	540	600
$p_{\text{tot}} / \text{arb.}$	110	120	129	137	144	148	156	159	164	167

The initial pressure was 100 (arb. units)

- (b) Show that these data are consistent with a first-order rate law and obtain a value of the corresponding rate constant.
- (c) Similar measurements made at 319 °C and 308 °C gave first-order rate constants of $9.85 \times 10^{-4} \text{ s}^{-1}$ and $5.12 \times 10^{-4} \text{ s}^{-1}$, respectively. By plotting an appropriate graph, use these data, along the value obtained at 330 °C, to obtain a value of the activation energy and the pre-exponential factor.